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Course –ECA5601- 5g/6g Communication

Experiment: 6 5G NR Frame Structure Analysis with Different Subcarrier Spacing

Aim To analyze the 5G NR frame structure by studying the effect of different subcarrier spacings, slot durations, and OFDM symbols using MATLAB.

Objectives

1. To analyze different Subcarrier Spacing (SCS) values used in 5G NR numerology.
2. To compare the slot duration corresponding to different Subcarrier Spacing values in 5G NR.
3. To study the number of OFDM symbols present in a 5G NR slot for normal cyclic prefix.

Procedure

1. Define the MIMO system configuration (e.g., number of transmit and receive antennas such as 2×2 or 4×4).
2. Generate random binary data bits to represent the transmitted signal.
3. Modulate the bits using a digital modulation scheme (e.g., BPSK or QPSK).
4. Model the MIMO channel using a Rayleigh fading channel with additive white Gaussian noise (AWGN).
5. Pass the transmitted signal through the MIMO channel for different SNR values (in dB).
6. At the receiver, apply detection techniques (e.g., Zero-Forcing or Maximum Likelihood detection).
7. Demodulate the received signal to recover transmitted bits.
8. Compute Bit Error Rate (BER) by comparing transmitted and received bits.
9. Calculate Channel Capacity (bps/Hz) using SNR and MIMO capacity formula.
10. Plot BER vs SNR and Capacity vs SNR graphs for performance analysis.

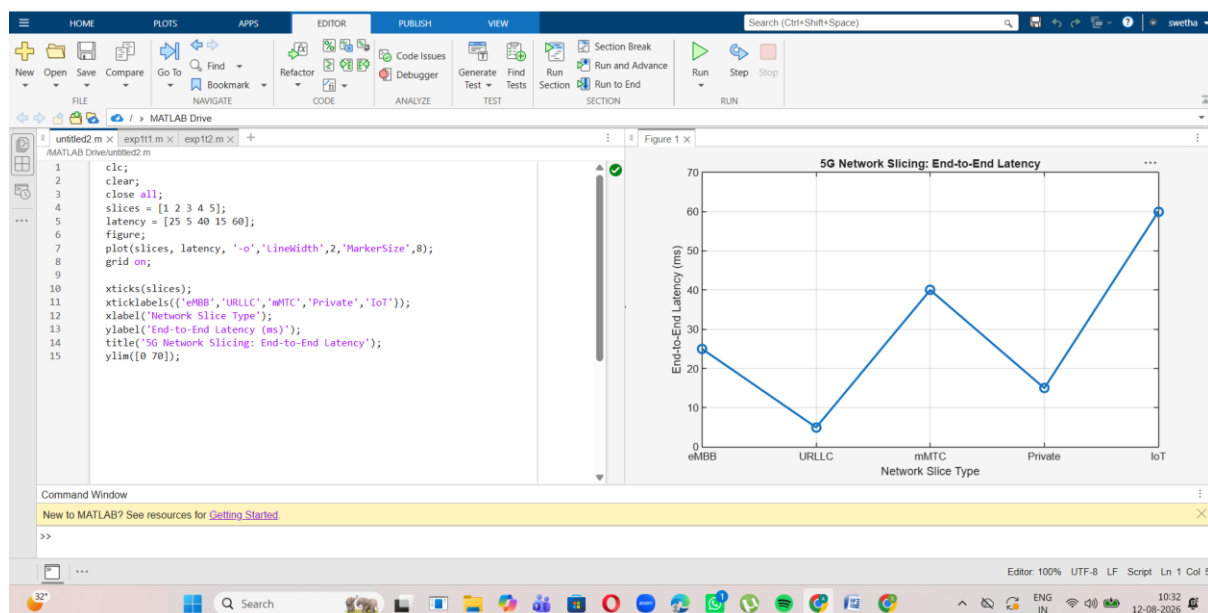
objective 1 :

Program

```
clc;
clear;
close all;
slices = [1 2 3 4 5];
latency = [25 5 40 15 60];
figure;
plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);
grid on;

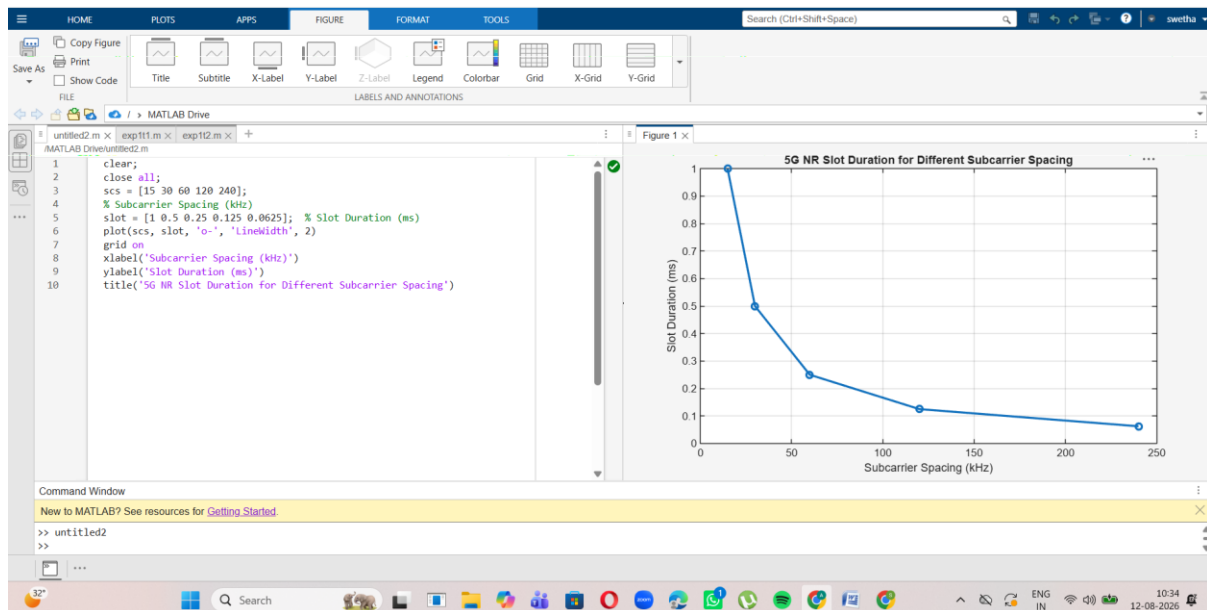
xticks(slices);
xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});
xlabel('Network Slice Type');
ylabel('End-to-End Latency (ms)');
title('5G Network Slicing: End-to-End Latency');
ylim([0 70]);
```

output



Objective 2: code and output:

```
clear;
close all;
scs = [15 30 60 120 240];
% Subcarrier Spacing (kHz)
slot = [1 0.5 0.25 0.125 0.0625]; % Slot Duration (ms)
plot(scs, slot, '-o', 'LineWidth', 2)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Slot Duration (ms)')
title('5G NR Slot Duration for Different Subcarrier Spacing')
```

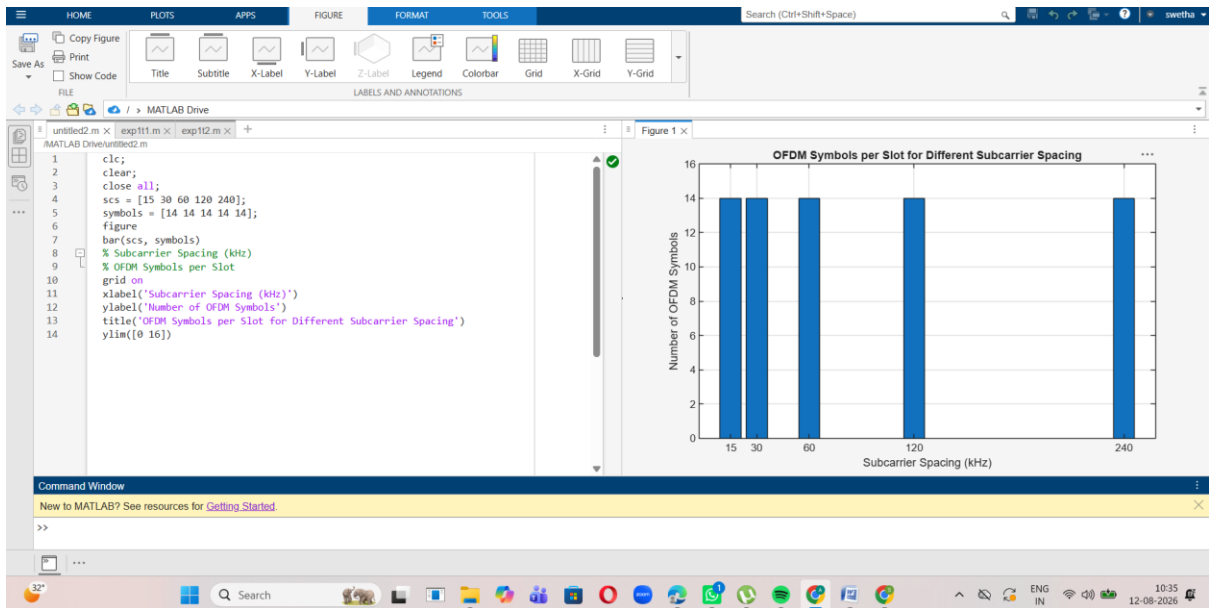


Objective 3:code and output:

```

clc;
clear;
close all;
scs = [15 30 60 120 240];
symbols = [14 14 14 14 14];
figure
bar(scs, symbols)
% Subcarrier Spacing (kHz)
% OFDM Symbols per Slot
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Number of OFDM Symbols')
title('OFDM Symbols per Slot for Different Subcarrier Spacing')
ylim([0 16])

```



Result:

The MATLAB analysis successfully demonstrated the flexible numerology of 5G NR. The results showed that increasing the subcarrier spacing decreases the slot duration while maintaining a constant number of OFDM symbols per slot under the normal cyclic prefix configuration.